ECO 2905 Assignment (Fall 2026, due date: Oct 20)

1 Uber Experiment

We conducted an Uber experiment in class. The data file has been uploaded to Quercus.

(5 points) Download the file from Quercus and extract the data from the PDF into a format suitable for analysis.

(10 points) Examine the data and identify as many interesting patterns as you can. For each pattern, clearly describe what you find and provide a plausible economic explanation.

(5 points) Based on your findings, propose a small randomized controlled trial (RCT) that could be conducted using the Uber app. Suppose 200 students are willing to participate. Clearly describe:

- the treatment and control groups;
- the outcome(s) you would measure; and
- the economic question the experiment would help answer.

2 Satellite Images

In this part of the assignment, you will build and train a neural network to identify residential houses using a subset of our labeled satellite-image data. There are no restrictions on the architecture, training procedure, or computing device you use. The purpose of the exercise is to give you hands-on experience with applying a machine-learning algorithm and dealing with the practical issues that arise in training and evaluating a model.

We will provide cropped property images from Toronto satellite imagery. Your task is to train a classifier that labels each image as either **H** or **A**. An image receives label **H** if it represents a residential house according to the labeling standard described below; otherwise, it receives label **A**.

Because each image is created by overlaying satellite imagery with property-boundary information, a residential house may be split across multiple satellite images. As a result, a cropped image may contain only a small portion of a house or nearly the entire structure. Under our labeling rule, an image is labeled **H** if the house has a non-flat roof and a clearly visible portion of that roof appears in the image. We recognize that this criterion is somewhat subjective, so you should use the labeled training images and the examples in Section 3 to understand the intended standard.

We provide the following files, which you can find in this folder CLICK:

- `TrainData.zip`: training images, including 4,000 images labeled **H** and 2,400 images labeled **A**;
- `TrainData.xlsx`: the corresponding labels for `TrainData`;
- `SupplementData.zip`: 800 additional unlabeled images that you may use if helpful;
- `B16_L4_S2_S.pth`: pretrained ResNet-50 weights for this classification task; and
- `House_Lables.ipynb`: sample code for running our pretrained model.

2.1 Run the Pretrained Model (5 points)

Download the provided files and unzip `TrainData`. Keep the image folder, label file, model weights, and sample notebook in the same working directory.

Run the sample code and report the two confusion matrices produced in the notebook. You are responsible for resolving any package or environment compatibility issues that arise. Handling such issues is a common part of applied computational work.

2.2 Build Your Own Model (10 points)

Use `TrainData` to build and train your own classification model.

1. **Describe the architecture you use.** You may design your own architecture or use an existing network. For example, you may use a pretrained ResNet-18 from the `torchvision` package. If you design your own architecture, briefly describe its structure, including the number of layers, the number of channels in each layer, and the kernel sizes used. If you choose ResNet-50, you may not initialize it using the pretrained weights that we provide.
2. **Describe your training procedure.** At a minimum, report the following:
   - how you split `TrainData` into training and validation sets;
   - the main hyperparameters you tune and how you choose their values;
   - the loss function you use and why it is appropriate;
   - how you select your best model (for example, using validation loss or another validation metric);
   - how many training runs you complete, how many epochs are used in each run, and what criterion you use to determine convergence; and
   - the confusion matrix for your best model evaluated on the validation set.
3. **(Optional) Label part or all of `SupplementData` yourself and retrain your model.** Report the confusion matrix for the updated model. You do not need to label all 800 images; use as many as you believe are useful. The purpose of this exercise is to explore how additional data and label quality affect model performance. You may use the newly labeled observations as additional training data or for supplementary validation exercises. However, when comparing your original and updated models, evaluate both on the **same validation set from `TrainData`**, leaving its images and labels unchanged.

2.3 Bonus (5 points)

After you submit your final model, we will release an additional 800 labeled images to serve as a test set. You may then compare your model's performance with our pretrained model and with your classmates' models. Bonus credit will be based on performance on the test-set confusion matrix, with attention to both precision and recall for classes **H** and **A**.

2.4 What to Submit

Please submit the following:

- a report answering all required questions above and including the requested confusion matrices, as well as the optional experiment if you complete it;
- one notebook containing your training and evaluation code and the corresponding outputs, with settings consistent with those reported in your write-up; and
- the final model weights for your submitted model.

3 Appendix

The image below is labeled H:



Figure 1: House Example

The images below are labeled A:



Figure 2: Non-House Examples

The image below is labeled **H** because most of the roof is visible in the image:

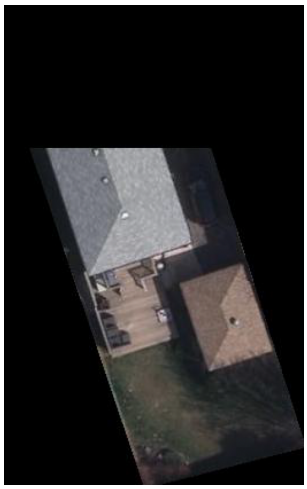


Figure 3: House Example

The image below is labeled **A** because only a small portion of the roof is visible:



Figure 4: Non-House Example